

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
		(iii)	Let the initial number of particles be N_0, so number of particles remaining after 57 days $= \frac{1}{32} N_0$ i.e. $A \propto N(1)$ Decrease in the number of nuclei in 57 days $= \left(1 - \frac{1}{32}\right) N_0$ i.e. decrease (1) Percentage decrease $= \frac{\left(1 - \frac{1}{32}\right) N_0}{N_0} \times 100\% = 96.875\%$ i.e. percentage (1) Answer only of 3% award 2 marks only	1	1 1		3	3	
			Question 1 total	7	4	0	11	7	4